

Monitoring, Drift Detection & Retraining Report

AIPI-561, Assignment 4

Overview

This week we investigate opportunities and challenges associated with event-driven or periodic data and concept drift. The former can silently drive performance deterioration and undermine confidence in a deployed system while the latter can motivate seemingly correct predictions. The backdrop here remains our NYC Taxi forecasting model, which induces demand based on historical measurements and a number of engineered features.

Monitoring Strategy

In this assignment we simulate the arrival of new data and establish a workflow to mine insights, monitor for drift, and trigger downstream actions like alerting a human user, blocking a commit, or executing a background retraining operation to attempt to recover when drift is detected. Monitoring occurs both with a sliding window at a 24-hour period and on commit to the repository.

Our sliding window is calibrated to allow enough data to accumulate to support stratified performance measurements. Shorter windows risk inadequate samples arriving to reason about performance on critical features. Longer windows start to risk business impact if we fail to detect an underlying problem. Commit-based monitoring allows us to validate new logic against known baselines and against data the deployed model has not yet been evaluated against. Concurrently the two strategies offer defense against two different sources of instability: real-world data and evolution of the system itself.

Metrics and Thresholds Employed

We settle on a handful of metrics that drive informational and system control flows. These are outlined below.

- **Mean Absolute Error (MAE):** Our primary modeling activity is forecasting the number of fares across rolling windows with respect to various geographic locations. While real trip counts are integer values, our decision to use regression results in a continuous value prediction. While this could be binned and compared with ground truth to compute an accuracy score, we elect to use a more natural and interpretable metric for regression with MAE. We monitor raw predictions on 15 minutes of aggregate data for MAE that exceeds 150% of the baseline MAE. This is applied both across ordinal zone values and for each zone. The latter is a hedge

against asymmetric deterioration of performance which could be hidden by use of exclusively aggregate statistics.

- Null Rates: We perform a simple check on null rates to ensure our upstream sensors and feeds continue to operate as expected.
- Standard Deviation: We examine the standard deviation of our target value to help flag potential concept drift which would notionally, although we don't implement it here, trigger a retraining decision.
- Kolmogorov-Smirnov (KS) Test: We employ the KS test as a supplemental statistic to aid in detecting distribution shift from our baseline dataset. This is a compact and powerful statistic that measures the largest distance between two cumulative distribution functions, but it must be interpreted carefully and calibrated to the application. We elect to use the KS statistic itself (D) in lieu of the associated p-value because of the high risk of false positives in production systems with the latter. We alarm at $D > 0.1$ which was empirically determined based on historical data.
- Duplicate Rate: We monitor the ratio of duplicate rows in our new dataset and report the resulting metric for human review.

Reflection

In general, the concept of monitoring a deployed system to ensure assumptions hold is a straightforward concept. However the iterative process of codifying these assumptions, devising appropriate tests to detect when they are violated, and taking downstream actions is non-trivial. In the end this assignment reinforces the notion that without a proper appreciation of the tributary data sources, individual features and resulting system outputs, we are unlikely to preserve system performance for long. A rigorous approach is needed to guard against the erosion of system integrity and maintain confidence in deployed models.

Appendix – Automation Screenshots

monitor

failed 1 hour ago in 42s

Q Search logs

🔄 ⚙️

> Set up job

1s

> Run actions/checkout@v3

13s

> Set up Python

0s

> Install dependencies

17s

> Compute monitoring metrics

2s

> Detect drift patterns

4s

1 ▶ Run python -m week4.scripts.detect_drift

11 =====

12 DRIFT DETECTION

13 =====

14 Loading datasets...

15 Extracting last 24 hours (2026-02-27 23:45:00 -> 2026-02-28 23:45:00)

16 Loading model...

17 Loading model...

18 Loading model...

19 Loading model...

20

21 =====

22 DRIFT DETECTION REPORT

23 =====

24

25 Weekend Drift

26 -----

27 Drift detected : True

28 KS statistic : 0.7333

29 p-value : 0.000e+00

30

31 Trip Drift

32 -----

33 Drift detected : False

34 KS statistic : 0.0656

35 p-value : 1.279e-19

36

37 Hour Drift

38 -----

39 Segments checked: 24

40 Segments drifted: 12

41

42 segment base_mae new_mae delta ratio drift

43 -----

44 10 3.24 7.04 3.80 2.17 True

45 17 4.36 6.88 2.52 1.58 True

46 19 4.43 6.95 2.52 1.57 True

47 9 2.88 5.40 2.52 1.87 True

48 1 1.59 4.11 2.51 2.58 True

49 22 3.72 6.21 2.49 1.67 True

50 23 3.25 5.53 2.29 1.70 True

51 0 2.53 4.61 2.08 1.82 True

52 12 3.53 5.50 1.96 1.56 True

53 2 1.28 3.12 1.84 2.43 True

54

55 Loc Drift

56 -----

57 Segments checked: 57

58 Segments drifted: 25

59

60 segment base_mae new_mae delta ratio drift

61 -----

62 79 3.35 8.82 5.47 2.63 True

63 249 3.42 8.10 4.68 2.37 True

64 237 5.36 9.66 4.30 1.80 True

65 236 4.84 8.77 3.93 1.81 True

66 239 4.09 7.69 3.59 1.88 True

67 161 5.25 8.80 3.56 1.68 True

68 230 5.10 8.44 3.34 1.65 True

69 48 4.03 7.15 3.12 1.78 True

70 234 3.66 6.73 3.07 1.84 True

71 114 2.69 5.54 2.86 2.06 True

72 Error: Process completed with exit code 1.

Figure 1 – Github automation reporting drift statistics and failing associated action

is:issue state:open

LabelsMilestonesNew issue

☐ Open 8 Closed 0

AuthorLabelsProjectsMilestonesAssigneesNewest

☐ Drift detected in monitoring drift-alert
#8 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#7 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#6 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#5 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#4 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#3 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#2 · github-actions opened 1 hour ago

☐ Drift detected in monitoring drift-alert
#1 · github-actions opened 1 hour ago

Figure 2 – Automatic issue creation on drift detection